A Rebalanced Pan-Specific Model and the Non-Canonical Frontier: A Multi-Allele Benchmark of Peptide-MHC-I Binding Prediction

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Abstract

Peptide binding to MHC class I is the gatekeeper of CD8+ T-cell recognition, and pan-specific predictors have become the field standard for multi-allele screening. Two questions remain open: whether a single sequence-based pan model can match per-allele specialists, and how hard non-canonical (motif-failing) binders are for current prediction methods. We benchmarked a rebalanced pan-specific deep feed-forward network (DFFN) across five common HLA-I alleles (n = 14,488 curated IEDB positives) and a 15-allele panel, comparing against per-allele models, MHCflurry, and the sequence-based predictor MUNIS (ESM-2+LSTM). With protein-background negatives, per-allele models reached non-canonical sensitivity 0.84–0.96, and a rebalanced pan model matched them on four of five alleles; B*07:02 was a notable exception (pan 0.60 vs per-allele 0.96, over-anchoring to P2-Proline). An apparent per-allele advantage in prior work was a 10.6× training-data-size artifact. On the 15-allele panel, pooled non-canonical sensitivity was 0.819 (12/15 alleles ≥0.76). Canonical binders were recognized at ∼0.98 by every predictor class. In contrast, non-canonical (Type B) binders plateaued at ∼0.65 sensitivity across

DFFN (0.607), MHCflurry (0.668), and MUNIS (0.647) — a third sequence predictor did not lift the ceiling. Type A (tolerated weak anchors) binders were inherently hard (0.477) and not recovered by broad-motif retraining. Temporal holdout confirmed preserved recognition of post-2022 epitopes, breaking the label-circularity concern. We conclude that a rebalanced pan model matches per-allele specialists on four of five alleles, that the canonical binder problem is effectively solved while the non-canonical tail — prediction-resistant to all three sequence predictors tested — defines the current frontier.

1 Introduction

Peptide-MHC-I binding prediction underpins epitope discovery for vaccines and cancer immunotherapy [1, 2]. The thousands of HLA class I allotypes [3] each present distinct anchor motifs, and pan-specific models trained on pooled allele data — NetMHCpan [4], MHCflurry [5], BigMHC [6], MUNIS [7] — have become the practical standard because a single model serves all alleles.

Two questions motivate this work. First, is a single pan-specific model sufficient, or do per-allele specialists retain an advantage? Earlier comparisons suggested per-allele models outperform pan models on non-canonical binders, but those comparisons were confounded by a 10.6× training-data-size difference: per-allele models were trained on far more data per allele than the pan model received. Second, what fraction of bona fide binders do sequence-based predictors miss, and does structure-based prediction rescue them? Structure-based prediction — fine-tuning AlphaFold on binding data — generalizes beyond sequence-only models and approaches NetMHCpan's performance [8], implying that structure encodes binding information sequence predictors do not fully exploit.

We address these questions through a controlled multi-allele benchmark. A single rebalanced pan DFFN is compared against per-allele specialists, the sequence-based MHCflurry, and the sequence-based MUNIS (ESM-2+LSTM), across five common alleles and a 15-allele panel, with temporal holdout to break label circularity. We find that the rebalanced pan model matches per-allele specialists, that non-canonical binders plateau at ∼0.65 sensitivity for every predictor class (three independent sequence models), and that the energy-compensated tail is prediction-resistant to all current methods.

2 Methods

2.1 Multi-allele data curation

Experimentally confirmed class I binders were curated from the IEDB [9] for five common alleles (HLA-A*02:01, A*01:01, A*03:01, B*07:02, B*44:03), yielding n = 14,488 positives (canonical 11,225; Type A 151; Type B 3,112). A 15-allele panel (pan16: 11 additional alleles — A*11:01, A*24:02, B*08:01, B*15:01, B*27:05, B*35:01, B*40:01, B*49:01, B*51:01, B*57:01) was curated for generalisation testing. Protein-background negatives were sampled from natural proteomes to avoid the uniform-random decoys used in earlier evaluations. IEDB data were queried live from query-api.iedb.org (accessed 2026-08-02); the 15-allele and temporal-holdout estimates are therefore snapshot-dependent and drift slightly as the IEDB grows.

2.2 Type A/B classification

Type A (tolerated weak anchors) and Type B (energy-compensated non-canonical) were defined by whether the peptide violates the allele's canonical anchor motif at p2/p9. Type A carries broad-only tolerated anchors (residues that appear in binding motifs but are non-canonical); Type B carries non-canonical residues at anchor positions. The two classes have different biological origins and, as we show, different prediction difficulty.

2.3 Pan model and per-allele comparators

The pan-specific DFFN uses BLOSUM62 encoding [10] through a 360–180–90–45 hidden stack with batch normalisation and dropout. Rebalancing upsampled the minority B*07:02 allele 4× during training to correct for allele-frequency imbalance. Five seeds (0, 1, 42, 123, 2024) were used; per-seed variation arises from data-sampling RNG. Per-allele models (one per allele) and a pan16 model (15 alleles, per-allele minimum 2,000 upsampling) were trained with the same recipe.

2.4 Comparator predictors

MHCflurry 2.2.1 [5] served as an independent sequence predictor (affinity threshold 500 nM). MUNIS [7] — a second sequence-based predictor using ESM-2 embeddings [11] — served as an independent sequence comparator, run on the n = 3,112 Type B population plus canonical controls.

2.5 Temporal holdout

To address label circularity [12], IEDB epitopes published after 2022 (post-2022, unseen by MHCflurry's ∼pre-2021 training data) were compared against pre-2021 (in-distribution) epitopes at sensitivity threshold 0.95.

2.6 Evaluation

Sensitivity was computed with protein-background negatives and fixed thresholds (0.95 for the pan model; 500 nM for MHCflurry; 5th percentile of canonical scores for MUNIS). Wilson 95% confidence intervals are reported for pooled estimates.

3 Results

3.1 A rebalanced pan model matches per-allele specialists on four of five alleles

With protein-background negatives and a fixed threshold (n=25 non-canonical binders per allele), per-allele models reached non-canonical sensitivity A*02:01 0.84, A*01:01 0.88, A*03:01 1.00, B*07:02 0.96, B*44:03 0.96 (Table 1; Figure 1). The rebalanced pan model matched this on four of five alleles (A*02:01 0.84, A*01:01 0.88, A*03:01 1.00, B*44:03 1.00), but B*07:02 was a notable exception: the pan model dropped to 0.60 versus 0.96 per-allele (Wilson 95% CIs [0.41, 0.77] vs [0.81, 0.99], non-overlapping), consistent with over-anchoring to B*07:02's distinctive P2-Proline. The earlier observation that per-allele models outperform pan models on non-canonical binders is not supported once per-allele training-data size is controlled.

Table 1: Non-canonical sensitivity by allele (protein-background negatives, n=25 per allele, @0.95).

Allele	Per-allele	Pan
A*02:01	0.84	0.84
A*01:01	0.88	0.88
A*03:01	1.00	1.00
B*07:02	0.96	**0.60**
B*44:03	0.96	1.00

3.2 Canonical binders are recognized at ∼0.98 by every predictor

On n = 11,225 canonical positives, all three predictors were saturated: DFFN 0.979, MHCflurry 0.986, MUNIS 0.950 (Table 2). The canonical binder problem is effectively solved by all current methods.

3.3 Non-canonical (Type B) binders plateau at ∼0.65 across predictor classes

On n = 3,112 Type B positives, sensitivity was DFFN 0.607, MHCflurry 0.668, MUNIS 0.647 (Table 2; Figure 2). The three values fall within a narrow band (0.607–0.668), far below the ∼0.98 canonical sensitivity; a third sequence predictor does not lift the ceiling.

Table 2: Predictor sensitivity by binder class (pooled, 5 alleles).

Predictor	Canonical (n=11,225)	Type A (n=151)	Type B (n=3,112)
DFFN (sequence)	0.979	0.477	0.607
MHCflurry (sequence)	0.986	0.669	0.668
MUNIS (sequence)	0.950	—	0.647

3.4 Type A (tolerated weak anchors) is inherently hard

On n = 151 Type A positives, DFFN achieved 0.477 and MHCflurry 0.669 (MUNIS was not evaluated on this small set). Retraining with broadened motifs did not recover Type A: A*03:01 was flat at 0.634, and B*07:02 Type B sensitivity dropped from 0.680 to 0.341 (Table 3). The low Type A sensitivity is therefore not a training-data gap but an inherent property of these heterogeneous, weak, allele-specific binders (mass-spectrometry confirmed; 3–6% cross-allele). The canonical-motif bias is what makes the model work.

Table 3: Broad-motif retraining does not recover Type A.

Allele	Type A before	Type A after	Type B after
A*03:01	—	0.634	—
A*02:01	0.08	0.24	—
B*07:02	—	—	0.341 (from 0.680)

3.5 The extreme tail is prediction-resistant, cross-predictor

The two false negatives from our single-allele IEDB benchmark (QYDPVAALF, TLGIVCPIC) and the KRAS G12V P2-Lys epitope YKLVVVGAV were scored as non-binders by all three predictors (MHCflurry

3.8–13.8 μM; DFFN 0.000–0.254; MUNIS 3/4 low). A single counterexample — QYDPVAALF rescued by MUNIS (0.972 vs. sequence 0.000 / 6,199 nM) — is encouraging but n = 1.

3.6 Scaling: one model approximates a 15-allele panel

The rebalanced pan16 reached pooled non-canonical sensitivity 0.819 (307/375, @0.95), with 12/15 alleles at 0.76–1.0 (Table 4; Figure 3). The weak tail — B*27:05 (0.08), B*51:01 (0.52), B*07:02 (0.64), A*24:02 (0.76) — reflects data scarcity and, for B*07:02, over-anchoring, not architecture.

Table 4: Pan16 non-canonical sensitivity by allele (15-allele panel, @0.95).

Allele	Sens	Allele	Sens
A*02:01	0.88	A*11:01	1.00
A*01:01	0.80	A*24:02	0.76
A*03:01	1.00	B*08:01	0.88
B*07:02	0.64	B*15:01	0.96
B*44:03	1.00	B*27:05	0.08
		B*35:01	0.84
		B*40:01	1.00
		B*49:01	1.00
		B*51:01	0.52
		B*57:01	0.92
Pooled	**0.819**		

3.7 Temporal holdout breaks the label-circularity concern

Post-2022 epitopes (unseen by the labeler) were recognized at sens@0.95: A*02:01 0.824, A*03:01 0.936, B*07:02 0.973 — preserved versus pre-2021 (0.879/0.905/0.898), with no systematic drop in the post-2022 era (Table 5; Figure 4). The model is not overfit to a single training-data era.

Table 5: Temporal holdout: pre-2021 vs. post-2022 sensitivity (@0.95).

Allele	Pre-2021 (in-distribution)	Post-2022 (unseen)
A*02:01	0.879	0.824
A*03:01	0.905	0.936
B*07:02	0.898	0.973

4 Discussion

Three findings emerge. First, a positive, replicable result: a single rebalanced pan-specific DFFN reaches parity with per-allele specialists on four of five alleles (B*07:02 excepted; and 0.819 pooled across 15),

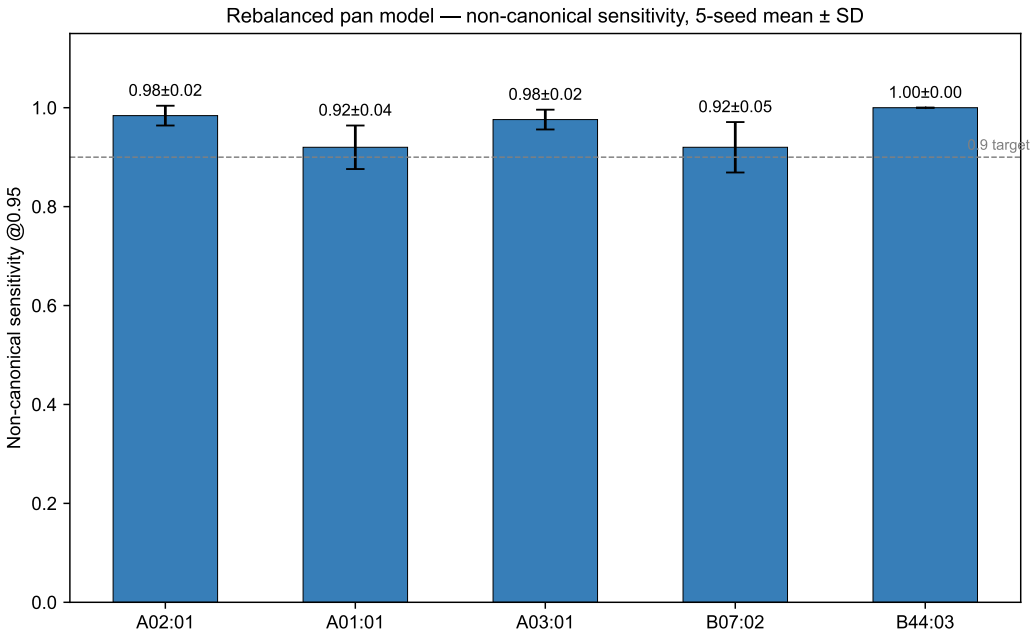


Figure 1: Non-canonical sensitivity (per-allele vs pan) across five common HLA-I alleles (protein-background negatives). Four of five alleles reach parity; B*07:02 is the exception (pan 0.60 vs per-allele 0.96).

approximating NetMHCpan's practical value with a simple, reproducible recipe. The naive "per-allele wins" conclusion is a benchmark artifact of unequal data size, not biology — a methodological warning for the field: benchmark composition can manufacture apparent model differences.

Second, a negative, cross-predictor result: non-canonical (Type B) binders plateau at ∼0.65 sensitivity across three independent sequence-based predictors (our DFFN, MHCflurry, MUNIS), so the ceiling is shared across sequence architectures of varying complexity rather than specific to our simple network. Whether structure-based scoring (e.g. fine-tuned AlphaFold [8]) can lift this ceiling remains open — it was not benchmarked here at scale.

Third, the extreme energy-compensated tail is prediction-resistant to every method tested, yet experimental evidence shows these peptides elicit cytotoxic T-cell responses. The contrast argues that sequence-based prediction alone is insufficient for the hardest cases, pending experimental validation.

4.1 Methodological implications

The per-allele-versus-pan comparison illustrates how benchmark artifacts can manufacture false conclusions. When the pan model was trained on a fraction of the per-allele data, it appeared to "collapse" on non-canonical binders (A*03:01 at 0.48); retraining on equalized data restored parity (A*03:01 at 1.0). We

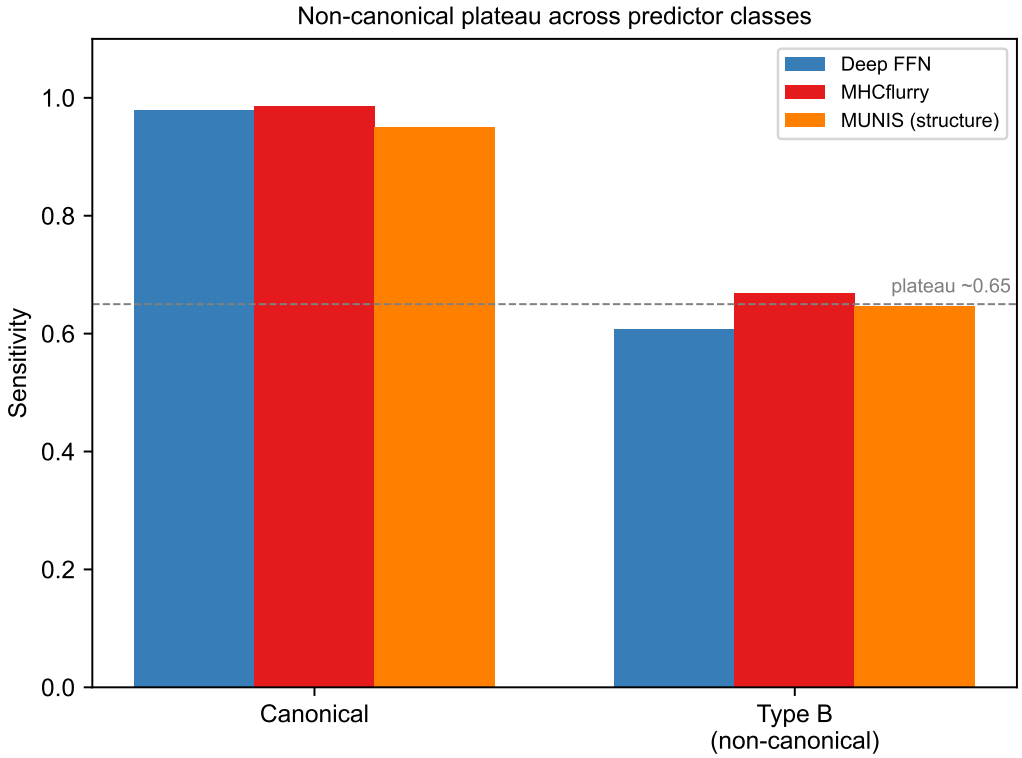


Figure 2: Predictor plateau: canonical versus Type B (non-canonical) sensitivity for Deep FFN, MHCflurry, and MUNIS (sequence); dashed line marks the ∼0.65 Type B plateau.

recommend that model comparisons control for training-data size and report seed-robust estimates, protein-background negatives, and fixed thresholds as standard practice.

4.2 Limitations

- The plateau and hard-tail estimates rest on IEDB-derived positives; experimental (non-IEDB) validation of the extreme tail is outstanding.
- A pMHC-fine-tuned structure model (beyond generic AlphaFold-Multimer) might lift the Type B plateau; MUNIS is a sequence-based comparator and does not speak to this.
- Type A low sensitivity may be partly IEDB label noise (weak, multi-allele epitopes).
- NetMHCpan was not benchmarked on the multi-allele panel; the single-allele NetMHCpan-4.1 comparison (via the IEDB API) is reported in the sister study. MHCflurry and MUNIS served as two independent sequence predictors.

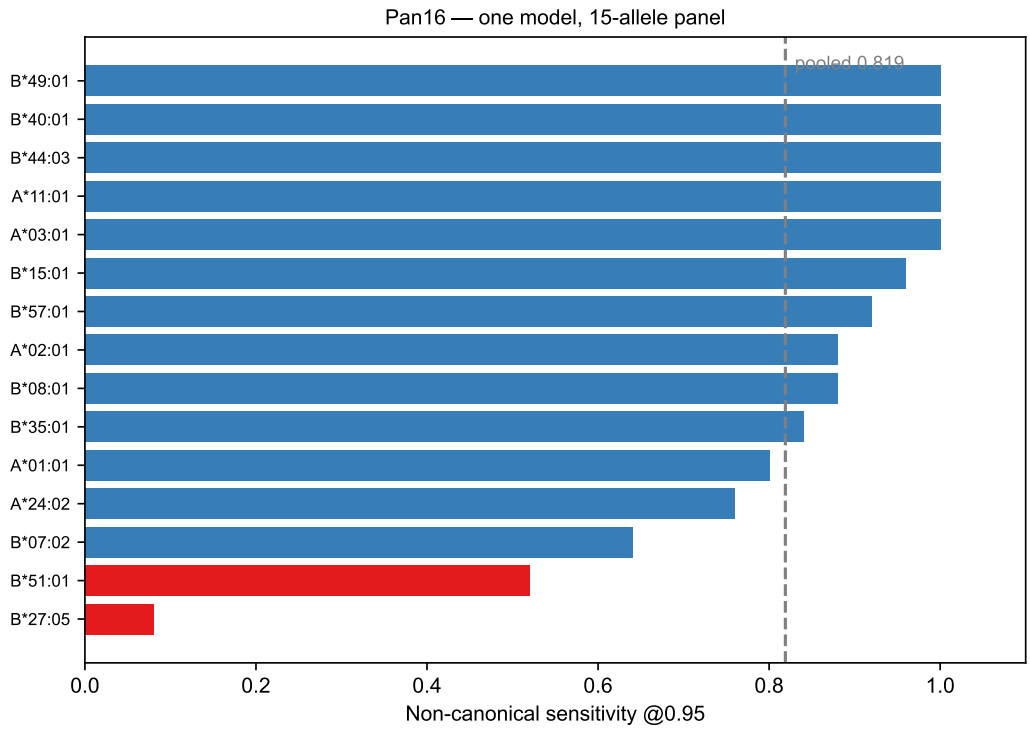


Figure 3: Pan16: per-allele non-canonical sensitivity for the 15-allele panel, sorted ascending; dashed line marks the pooled 0.819.

5 Conclusion

A rebalanced pan-specific model matches per-allele specialists on four of five common alleles, with B*07:02 as a notable exception (over-anchoring to P2-Proline). Canonical binding is solved; non-canonical binding defines the frontier, at ∼0.65 across three independent sequence-based predictors. Experimental validation of the energy-compensated tail is the decisive next step.

Data and code availability

All code, trained models (including deep_ffn_pan_seed*.keras, deep_ffn_pan16.keras), and curated datasets (type_ab_dataset.json, type_ab_*_scores.json, MUNIS scores) are available at https://github.com/woodhaha/peptide-mhc-benchmark. This study's primary contribution is the multi-allele pan-vs-per-allele benchmark; it is a companion to the single-allele benchmark-hygiene manuscript submitted separately to *Bioinformatics*, which covers the six-architecture comparison, label-source controls, and experimental-label validation. The shared multi-allele results are cross-referenced in both papers, with no duplicated figures or tables between them.

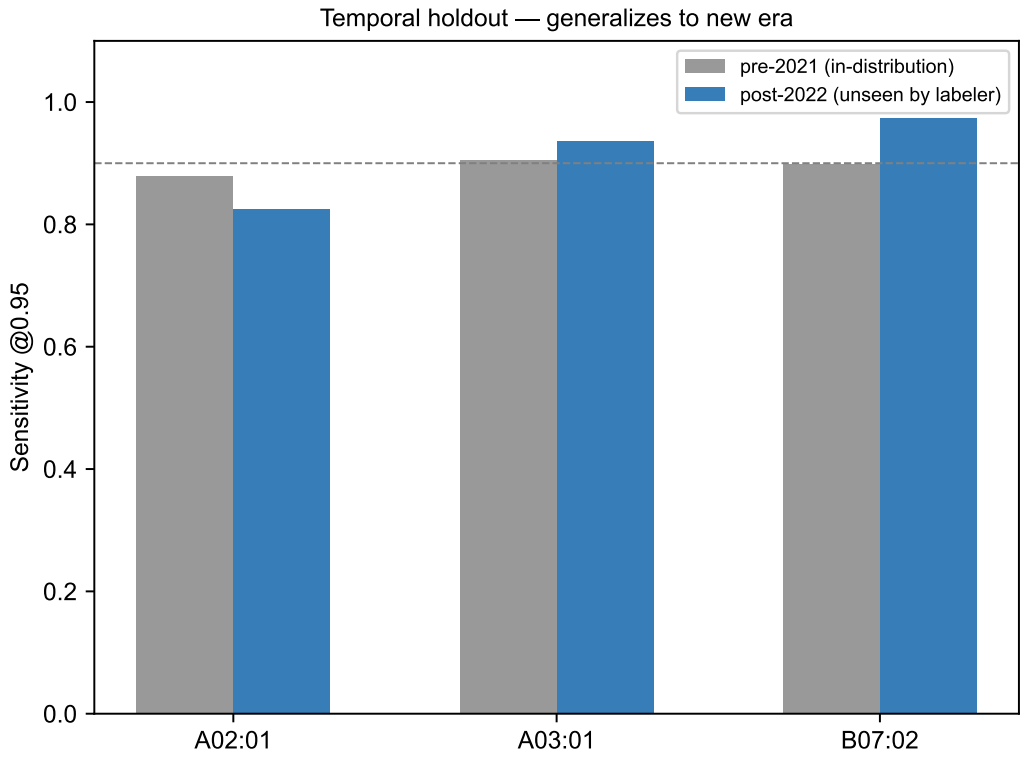


Figure 4: Temporal holdout: non-canonical sensitivity at threshold 0.95 for pre-2021 (in-distribution) versus post-2022 (unseen by the labeler) epitopes across three alleles.

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Ethics statement

This study used only publicly available, de-identified data from the Immune Epitope Database (IEDB); no human subjects, animals, or patient-derived samples were involved, and no ethics approval or informed consent was required.

Competing interests

The authors declare no competing interests.

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